

Assignment 3 — Spark vs Ray Distributed Data Pipeline

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October 5, 2026

1 Overview

This document describes the procedure used to implement an identical NYC Yellow Taxi cleaning and transformation pipeline in both **Apache Spark (PySpark 3.5.1)** and **Ray (2.59.0, Ray Data)**, run each on a distributed cluster, and benchmark execution time, resource utilization, and Python UDF overhead.

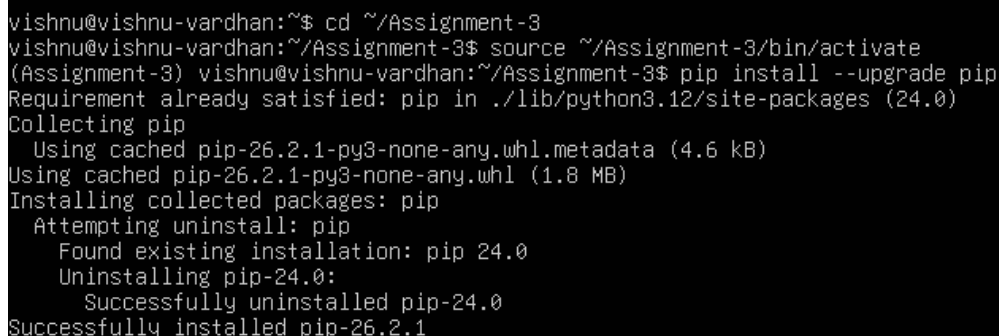
2 Environment Setup

2.1 System Prerequisites

```
sudo apt update
sudo apt install -y openjdk-17-jdk python3-pip python3-venv
```

2.2 Python Virtual Environment

```
python3 -m venv ~/Assignment-3
source ~/Assignment-3/bin/activate
pip install --upgrade pip
```



```
vishnu@vishnu-varadhan:~$ cd ~/Assignment-3
vishnu@vishnu-varadhan:~/Assignment-3$ source ~/Assignment-3/bin/activate
(Assignment-3) vishnu@vishnu-varadhan:~/Assignment-3$ pip install --upgrade pip
Requirement already satisfied: pip in ./lib/python3.12/site-packages (24.0)
Collecting pip
  Using cached pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.2.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 24.0
    Uninstalling pip-24.0:
      Successfully uninstalled pip-24.0
Successfully installed pip-26.2.1
```

Figure 1

2.3 Install Frameworks

```
pip install pyspark==3.5.1
pip install "ray[default]==2.59.0" pandas pyarrow
```

```
(Assignment-3) vishnu@vishnu-varadhan:~/Assignment-3$ python -c 'import pyspark, ray; print("both imported OK")'
both imported OK
(Assignment-3) vishnu@vishnu-varadhan:~/Assignment-3$ java --version
openjdk 17.0.20.1 2026-08-18
OpenJDK Runtime Environment (build 17.0.20.1+1-1-24.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 17.0.20.1+1-1-24.04-Ubuntu, mixed mode, sharing)
```

Figure 2: Environment test.

2.4 Set SPARK_HOME

```
export SPARK_HOME=$(python -c "import pyspark, os; print(os.path.dirname(pyspark.__file__))")
export PATH=$SPARK_HOME/bin:$SPARK_HOME/sbin:$PATH
echo $SPARK_HOME
```

3 Data Acquisition

```
mkdir -p ~/taxi_data/raw ~/taxi_data/processed
cd ~/taxi_data/raw

wget https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-01.parquet
wget https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv

ls -lh
```

Expected output:

```
yellow_tripdata_2024-01.parquet 48M
taxi_zone_lookup.csv 13K
```

```
(Assignment-3) vishnu@vishnu-varadhan:~/Assignment-3$ cd ~/taxi_data/raw
(Assignment-3) vishnu@vishnu-varadhan:~/taxi_data/raw$ wget http://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-01.parquet
--2026-10-04 11:23:30-- http://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-01.parquet
Resolving d37ci6vzurychx.cloudfront.net (d37ci6vzurychx.cloudfront.net)... 18.67.153.6, 18.67.153.166, 18.67.153.210, ...
Connecting to d37ci6vzurychx.cloudfront.net (d37ci6vzurychx.cloudfront.net)|18.67.153.6|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-01.parquet [following]
--2026-10-04 11:23:30-- https://d37ci6vzurychx.cloudfront.net/trip-data/yellow_tripdata_2024-01.parquet
Connecting to d37ci6vzurychx.cloudfront.net (d37ci6vzurychx.cloudfront.net)|18.67.153.6|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 49961641 (48M) [binary/octet-stream]
Saving to: 'yellow_tripdata_2024-01.parquet'

yellow_tripdata_2024-01.parquet 100%[=====] 47.65M 3.01MB/s in 50s

2026-10-04 11:24:20 (980 KB/s) - 'yellow_tripdata_2024-01.parquet' saved [49961641/49961641]

(Assignment-3) vishnu@vishnu-varadhan:~/taxi_data/raw$ wget http://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv
--2026-10-04 11:26:05-- http://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv
Resolving d37ci6vzurychx.cloudfront.net (d37ci6vzurychx.cloudfront.net)... 99.86.148.201, 99.86.148.124, 99.86.148.12, ...
Connecting to d37ci6vzurychx.cloudfront.net (d37ci6vzurychx.cloudfront.net)|99.86.148.201|:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv [following]
--2026-10-04 11:26:05-- https://d37ci6vzurychx.cloudfront.net/misc/taxi_zone_lookup.csv
Connecting to d37ci6vzurychx.cloudfront.net (d37ci6vzurychx.cloudfront.net)|99.86.148.201|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12331 (12K) [text/csv]
Saving to: 'taxi_zone_lookup.csv'

taxi_zone_lookup.csv 100%[=====] 12.04K --KB/s in 0.002s

2026-10-04 11:26:05 (7.18 MB/s) - 'taxi_zone_lookup.csv' saved [12331/12331]

(Assignment-3) vishnu@vishnu-varadhan:~/taxi_data/raw$ ls -lh
total 48M
-rw-rw-r-- 1 vishnu vishnu 13K Feb 22 2024 taxi_zone_lookup.csv
-rw-rw-r-- 1 vishnu vishnu 48M Mar 21 2024 yellow_tripdata_2024-01.parquet
```

Figure 3: Data Collection

4 Spark Cluster Setup (2 Workers)

4.1 Configuration Files

```
mkdir -p $SPARK_HOME/conf
```

\$SPARK_HOME/conf/spark-env.sh:

```
export SPARK_MASTER_HOST=127.0.0.1
export SPARK_MASTER_PORT=7077
```

```
export SPARK_MASTER_WEBUI_PORT=8080
export SPARK_WORKER_CORES=1
export SPARK_WORKER_MEMORY=512m
export SPARK_WORKER_WEBUI_PORT=8081
export SPARK_DAEMON_MEMORY=100m
```

```
chmod +x $SPARK_HOME/conf/spark-env.sh
```

\$SPARK_HOME/conf/workers:

```
127.0.0.1
127.0.0.1
```

\$SPARK_HOME/conf/spark-defaults.conf:

```
spark.master spark://127.0.0.1:7077
spark.driver.memory 500m
spark.executor.memory 500m
spark.executor.cores 1
spark.sql.shuffle.partitions 4
spark.default.parallelism 2
spark.ui.port 4040
```

4.2 Start the Cluster

```
# Master
cd ~
nohup $SPARK_HOME/bin/spark-class org.apache.spark.deploy.master.Master \
  --host 127.0.0.1 --port 7077 --webui-port 8080 > spark-master.log 2>&1 &

# Worker 1
nohup $SPARK_HOME/bin/spark-class org.apache.spark.deploy.worker.Worker \
  spark://127.0.0.1:7077 --webui-port 8081 --cores 1 --memory 512m > spark-w1.log 2>&1 &

# Worker 2
nohup $SPARK_HOME/bin/spark-class org.apache.spark.deploy.worker.Worker \
  spark://127.0.0.1:7077 --webui-port 8082 --cores 1 --memory 512m > spark-w2.log 2>&1 &
```

4.3 Verify the Cluster

```
jps
```

Expected output:

```
Master
Worker
Worker
Jps
```

```
(Assignment-3) vishnu@vishnu-varadhan:~$ nohup $SPARK_HOME/bin/spark-class org.apache.spark.deploy.master.Master --host 127.0.0.1 --port 7077 --webui-port 8080 &
spark-master.log 2>&1 &
(1) 1929
(Assignment-3) vishnu@vishnu-varadhan:~$ jps
1971 Jps
1929 Master
(Assignment-3) vishnu@vishnu-varadhan:~$ nohup $SPARK_HOME/bin/spark-class org.apache.spark.deploy.worker.Worker spark://127.0.0.1:7077 --webui-port 8081 --cores
1 --memory 512m > spark-w1.log 2>&1 &
(2) 2000
(Assignment-3) vishnu@vishnu-varadhan:~$ jps
2050 Jps
2000 Worker
1929 Master
(Assignment-3) vishnu@vishnu-varadhan:~$ nohup $SPARK_HOME/bin/spark-class org.apache.spark.deploy.worker.Worker spark://127.0.0.1:7077 --webui-port 8082 --cores
1 --memory 512m > spark-w2.log 2>&1 &
(3) 2009
(Assignment-3) vishnu@vishnu-varadhan:~$ jps
2151 Jps
2000 Worker
2009 Worker
1929 Master
```

Figure 4: adding workers

```
curl -s http://localhost:8080 | grep -iE "Alive Workers|Cores|Status"
```

Expected output:

```
<li><strong>Alive Workers:</strong> 2</li>
<li><strong>Cores in use:</strong> 2 Total,
<li><strong>Status:</strong> ALIVE</li>
```

```
(Assignment-3) vishnu@vishnu-varadhan:~$ curl -s http://localhost:8080 | grep -iE "ALIVE|Workers|Cores" | head -10
<li><strong>Alive Workers:</strong> 2</li>
<li><strong>Cores in use:</strong> 2 Total,
<li><strong>Status:</strong> ALIVE</li>
<span class="collapse-aggregated-workers collapse-table" onClick="collapseTable('collapse-aggregated-workers','aggregated-workers')">
  <a>Workers (2)</a>
  <div class="aggregated-workers collapsible-table">
    <thead><th width="" class="">Worker Id</th><th width="" class="">Address</th><th width="" class="">State</th><th width="" class="">Cores</th><th width=""
class="">Memory</th><th width="" class="">Resources</th></thead>
    <tbody>
      <tr>
        <td>ALIVE</td>
        <td>ALIVE</td>
        <td><strong>Application ID</strong></td><td><strong>Name</strong></td><td><strong>Cores</strong></td><td><strong>Memory per Executor</strong></td><td><strong>Resources Per Executor</strong></td><td><strong>Submitted Time</strong></td><td><strong>User</strong></td><td><strong>State</strong></td><td>
width="" class="">Duration</td></tr></tbody>
```

Figure 5: Starting the cluster

4.4 Run the Spark Pipeline

```
python ~/spark_clean.py 2>&1 | tee ~/spark_run.log
```

```
(Assignment-3) vishnu@vishnu-varadhan:~$ python ~/spark_clean.py 2>&1 | tee ~/spark_run.log
26/10/04 17:31:32 WARN Utils: Your hostname, vishnu-varadhan resolves to a loopback address: 127.0.0.1; using 10.0.2.15 instead (on interface enp0s3)
26/10/04 17:31:32 WARN Utils: Set SPARK_LOCAL_IP if you need to bind to another address
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/10/04 17:31:36 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
[33.99s] Session started
[97.83s] Ingested: 2964624 trips, 265 zones
[115.02s] Cleansed: 2838136 trips
[166.06s] Joined: 2838136 rows
[220.28s] UDF+group done in 26.91s - 177 result rows
[291.45s] Exported parquet

=== SPARK TOTAL TIME: 291.45s ===
=== SPARK UDF TIME: 26.91s ===
```

Figure 6

```
[33.99s] Session started
[97.83s] Ingested: 2964624 trips, 265 zones
[115.02s] Cleansed: 2838136 trips
[166.06s] Joined: 2838136 rows
[220.28s] UDF+group done in 26.91s - 177 result rows
[291.45s] Exported parquet

=== SPARK TOTAL TIME: 291.45s ===
=== SPARK UDF TIME: 26.91s ===
```

Figure 7: Spark Result

4.5 Stop the Spark Cluster

```
killall -f "org.apache.spark.deploy.master.Master"
killall -f "org.apache.spark.deploy.worker.Worker"
sleep 3
jps
```

5 Ray Cluster Setup (2 CPUs)

5.1 Start the Ray Head Node with Dashboard

```
ray start --head --port=6379 --dashboard-host=0.0.0.0 \
--dashboard-port=8265 --num-cpus=2 --object-store-memory=500000000
```

```
(Assignment-3) vishnu@vishnu-varadhan:~$ ray start --head --port=6379 --dashboard-host=0.0.0.0 --dashboard-port=8265 --num-cpus=2 --object-store-memory=500000000
Usage stats collection is enabled. To disable this, add '--disable-usage-stats' to the command that starts the cluster, or run the following command: 'ray disable-usage-stats' before starting the cluster. See https://docs.ray.io/en/master/cluster/usage-stats.html for more details.

Local node IP: 10.0.2.15
2026-10-04 18:39:03,446 WARNING authentication_token_setup.py:73 -- Token authentication is disabled for this Ray cluster. Anyone with network access to the cluster can run arbitrary code and access its data. To enable token authentication, generate a token with 'ray get-auth-token --generate' or set RAY_AUTH_MODE=token. For more information, see https://docs.ray.io/en/latest/ray-security/token-auth.html
2026-10-04 19:00:06,285 ERROR services.py:1431 -- Failed to start the dashboard
2026-10-04 19:00:06,300 ERROR services.py:1456 -- Error should be written to 'dashboard.log' or 'dashboard.err'. We are printing the last 20 lines for you. See 'https://docs.ray.io/en/master/ray-observability/user-guides/configure-logging.html#logging-directory-structure' to find where the log file is.
2026-10-04 19:00:06,325 ERROR services.py:1500 --
The last 20 lines of /tmp/ray/session_2026-10-04-18-39-03_451937_4063/logs/dashboard.log (it contains the error message from the dashboard):

-----
Ray runtime started.
-----

Next steps
Note: The following commands are intended for use on
the head node or within the cluster network.

To add another node to this Ray cluster, run
  ray start --address='10.0.2.15:6379'

To connect to this Ray cluster:
  import ray
  ray.init()

To terminate the Ray runtime, run
  ray stop

To view the status of the cluster, use
  ray status
```

Figure 8

5.2 Verify the Cluster

```
sleep 20
ray status
```

```
(Assignment-3) vishnu@vishnu-varadhan:~$ ray status
===== Autoscaler status: 2026-10-04 18:01:12.364360 =====
Node status
-----
Active:
  1 node_7eaca82f4875d36349c33e14740ff142e9c656d1770e8ccb2176a175
Pending:
  (no pending nodes)
Recent failures:
  (no failures)

Resources
-----
Total Usage:
  0.0/2.0 CPU
  0B/2.84GiB memory
  0B/476.84MiB object_store_memory

From request_resources:
  (none)
Pending Demands:
  (no resource demands)
```

Figure 9: ray status

```
curl -s http://localhost:8265 | head -20
```

```
(Assignment-3) vishnu@vishnu-varadhan:~$ curl -s http://localhost:8265 | head -20
<!doctype html><html lang="en"><head><meta charset="utf-8"/><link rel="shortcut icon" href="/favicon.ico"/><meta name="viewport" content="width=device-width, in
itial-scale=1"/><title>Ray Dashboard</title><script defer="defer" src="/static/js/main.ff0259c7.js"></script><link href="/static/css/main.ec1426f1.css" rel="s
tylesheet"></head><body><noscript>You need to enable JavaScript to run this app.</noscript><div id="root"></div></body></html>(Assignment-3) vishnu@vishnu-varadh
```

Figure 10

5.3 Save Evidence Files

```
ray status > ~/ray_status_evidence.txt
curl -s http://localhost:8265 > ~/ray_dashboard_evidence.html
```

5.4 Run the Ray Pipeline

```
cd ~
python ~/ray_clean.py 2>&1 | tee ~/ray_run.log
```

```
[4.11s] Ray initialized
[7.33s] Ingested: 2964624 trips, 265 zones
[27.77s] Cleansed: 2838138 trips
[50.34s] Joined: 2838138 rows
[117.11s] UDF+group done in 62.14s - 177 result rows
[117.33s] Exported parquet

=== RAY TOTAL TIME: 117.33s ===
=== RAY UDF TIME: 62.14s ===
```

Figure 11: Ray result

6 Pipeline Parity Verification

```
python ~/run_all_checks.py 2>&1 | tee ~/parity_check.log
```

This script loads both output parquet files, joins them on (hour_of_day, Borough), and reports the maximum absolute difference in avg_speed_mph.

7 Results

Stage	Spark (s)	Ray (s)	Faster
Init	33.99	4.11	Ray
Ingest	63.84	3.22	Ray
Cleanse	17.19	20.44	Spark
Heavy Join	51.04	22.57	Ray
UDF + groupBy	26.91	62.14	Spark
Export	71.17	0.22	Ray
Total	291.45	117.33	Ray (2.48×)

Table 1: Per-stage execution time. Ray is 2.48× faster end-to-end.